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PTE 321 — Past Questions & Materials
by Renance

Course: PTE 321
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 400

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Paper 1 of 1 — PTE 321 - Petrochemical II

Level 400 • First Semester • Handout

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PTE 321 - Petrochemical II

Lecture 1: Petrochemicals: Overview

1.1 Introduction

-In this lecture, we present a brief overview of petrochemical technologies and discuss upon the general topology of the petrochemical process technologies.

-Petrochemicals refers to all those compounds that can be derived from the petroleum refinery products

-Typical feed stocks to petrochemical processes include

- C1 Compounds: Methane & Synthesis gas
- C2 Compounds: Ethylene and Acetylene
- C3 Compounds: Propylene
- C4 Compounds: Butanes and Butenes
- Aromatic Compounds: Benzene

-It can be seen that petrochemicals are produced from simple compounds such as methane, ethylene and acetylene but not multicomponent products such as naphtha, gas oil etc.

-1.1.1 Definition: These are the chemicals that are made from petroleum and natural gas. Petroleum and natural gas are made up of hydrocarbon molecules, which comprises of one or more carbon atoms, to which hydrogen atoms are attached.

-About 5 % of the oil and gas consumed each year is needed to make all the petrochemical products. Petrochemicals play an important role on our food, clothing, shelter and leisure. Because of low cost and easy availability, oil and natural gas are considered to be the main sources of raw materials for most petrochemicals.

1.1.2 Classification: Petrochemicals can be broadly classified into three categories:-

a.Light Petrochemicals: These are mainly used as bottled fuel and raw materials for other

organic chemicals. The lightest of these -- methane, ethane and ethylene -- are gaseous at room temperature. The next lightest fractions comprise petroleum ether and light naphtha

with boiling points between 80 and 190 degrees Fahrenheit.

b.Medium Petrochemicals: Hydrocarbons with 6-12 carbon atoms are called "gasoline"

which are mainly used as automobile fuels. Octane, with eight carbons, is a particularly good automobile fuel, and is considered to be of high quality. Kerosene contains 12 to 15 carbons and is used in aviation fuels, and also as solvents for heating and lighting.

c.Heavy Petrochemicals: These can be generally categorized as diesel oil, heating oil and

lubricating oil for engines and machinery. They contain around 15 and 18 carbon atoms with boiling points between 570 and 750 degrees Fahrenheit. The heaviest fractions of all are called "bitumens" and are used to surface roads or for waterproofing.

Bitumens can also be broken down into lighter hydrocarbons using a process called "cracking."

1.2 Process Topology

- Reactors: Reactors are the most important units in petrochemical processes. Petrochemicals are manufactured by following simple reactions using relatively purer feed stocks. Therefore, reaction chemistry for petrochemicals manufacture is very well established from significant amount of research in this field. Essentially all petrochemical processes need to heavily depend upon chemical transformation to first product the purification

-Separation: With distillation being the most important unit operation to separate the

unreacted feed and generated petrochemical product, the separation processes also play a major role in the process flow sheet. Where multiple series parallel

PETROCHEMICAL II (PTE 321)

el reactions are involved, the separation process assumes a distillation sequence to separate all products from the feed. A characteristic feed recycle will be also existent in the process topology. Apart from this, other separation technologies used in petrochemical processing units include phase separators, gravity settling units and absorption columns. Therefore, the underlying physical principle behind all these separation technologies is well exploited to achieve the desired separation.

-Dependence on Reaction pathway: A petrochemical can be produced in several ways

from the same feed stock. This is based on the research conducted in the process chemistry. For instance, phenol can be produced using the following pathways

- Peroxidation of Cumene followed by hydrolysis of the peroxide
- Two stage oxidation of Toluene
- Chlorination of Benzene and hydrolysis of chloro-benzene
- Direct oxidation of Benzene

-We can observe that in the above reaction schemes, there are two reaction pathways for phenol from benzene i.e., either chlorination of benzene or oxidation of benzene. Therefore, choosing the most appropriate technology for production is a trivial task.

Complexity in pathway: In the above Cumene example case, it is interesting to note that toluene hydrodealkylation produces benzene which can be used to produce phenol. Therefore, fundamentally toluene is required for the generation of various petrochemicals such as benzene and phenol. In other words, there is no hard and fast rule to say that a petrochemical is manufactured using a suggested route or a suggested intermediate petrochemical. Intermediate petrochemicals play a greater role in consolidating the manufacture of other downstream petrochemicals.



Figure 1: Petrochemical plant

Petrochemicals (also known as petroleum distillates) are chemical products derived from petroleum. Some chemical compounds made from petroleum are also obtained from other fossil fuels, such as coal or natural gas, or renewable sources such as corn, palm fruit or sugar cane.

The two most common petrochemical classes are:-

- 1-Olefins (including ethylene and propylene) and
- 2-Aromatics (including benzene, toluene and xylene isomers).

Oil refineries produce olefins and aromatics by fluid catalytic cracking of petroleum fractions. Chemical plants produce olefins by steam cracking of natural gas liquids like ethane and propane. Aromatics are produced by catalytic reforming of naphtha.

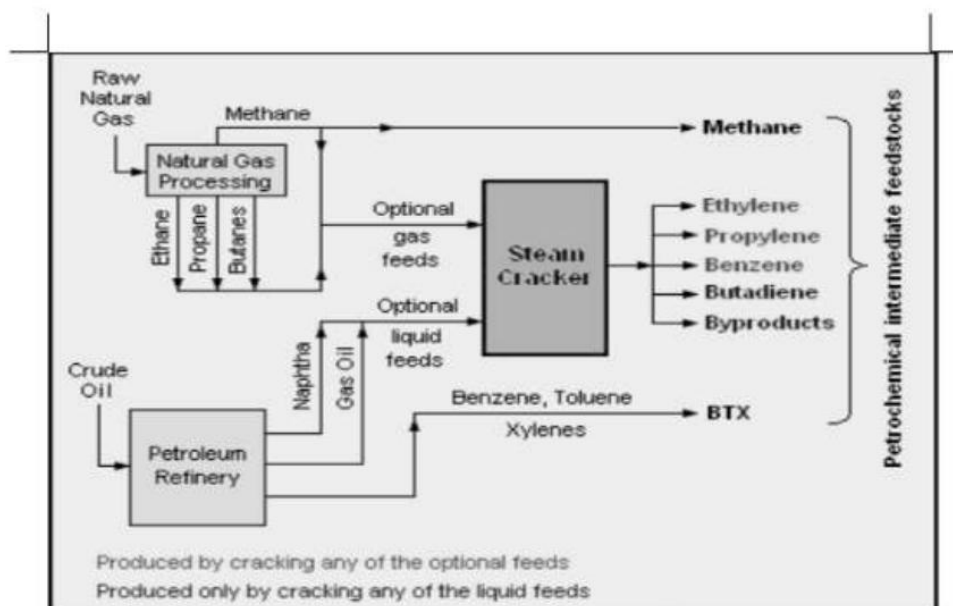
Olefins and aromatics are the building-blocks for a wide range of materials such as solvents, detergents, and adhesives. Olefins are the basis for polymers and oligomers used

in plastics, resins, fibers, elastomers, lubricants, and gels. [1121]

Global ethylene and propylene production are about 115 million tonnes and 70 million tonnes per annum, respectively. Aromatics production is approximately 70 million tonnes. The largest petrochemical industries are located in the USA and Western Europe; however, major growth in new production capacity is in the Middle East and Asia. There is substantial inter-regional petrochemical trade.

Primary petrochemicals are divided into three groups depending on their chemical structure:

- Olefins includes Ethene, Propene, Butenes and butadiene. Ethylene and propylene are important sources of industrial chemicals and plastics products. Butadiene is used in making synthetic rubber.
- Aromatics includes Benzene, toluene and xylenes, as a whole referred to as BTX and primarily obtained from petroleum refineries by extraction from the reformat produced in catalytic reformers using Naphtha obtained from petroleum refineries. Benzene is a raw material for dyes and synthetic detergent s, and benzene and toluene for isocyanates MDI and TDI used in making polyurethanes. Manufacturers use xylenes to produce plastics and synthetic fibers.
- Synthesis gas is a mixture of carbon monoxide and hydrogen used to make ammonia and methanol. Ammonia is used to make the fertilizer urea and methanol is used as a solvent and chemical intermediate. Steam crackers are not to be confused with steam reforming plants used to produce hydrogen and ammonia.



Primary Raw Materials for Petrochemicals:-

In general, primary raw materials are naturally occurring substances that have not been subjected to chemical changes after being recovered. Natural gas and crude oils are the basic raw materials for the manufacture of petrochemicals.

Secondary raw materials, or intermediates, are obtained from natural gas and crude oils through different processing schemes. The intermediates may be light hydrocarbon compounds such as methane and ethane, or heavier hydrocarbon mixtures such as naphtha or gas oil. Both naphtha and gas oil are crude oil fractions with different boiling ranges.

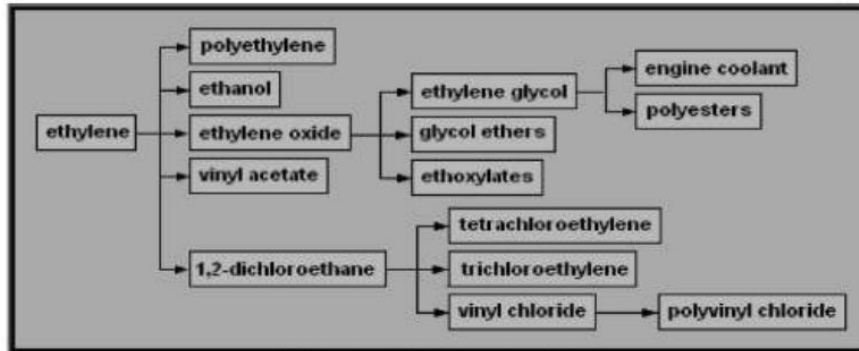
Coal, oil shale, and tar sand are complex carbonaceous raw materials and possible future energy and chemical sources. However, they must undergo lengthy and extensive processing before they yield fuels and chemicals similar to those produced from crude oils (substitute natural gas (SNG) and synthetic crudes from coal, tar sand and oil shale).

*ethylene:-

ethylene -the simplest olefin; used as a chemical feedstock and ripening stimulant

polyethylene -polymerized ethylene; LDPE, HDPE, LLDPE

- ethanol -via ethylene hydration(chemical reaction adding water)of
- ethylene oxide -via ethylene oxidation
- ethylene glycol -via ethylene oxide hydration
- engine coolant-ethylene glycol,water and inhibitor mixture
- polyesters -any of several polymers with ester linkages in the main chain glycol ethers -via glycol condensation
- ethoxylates
- vinyl acetate

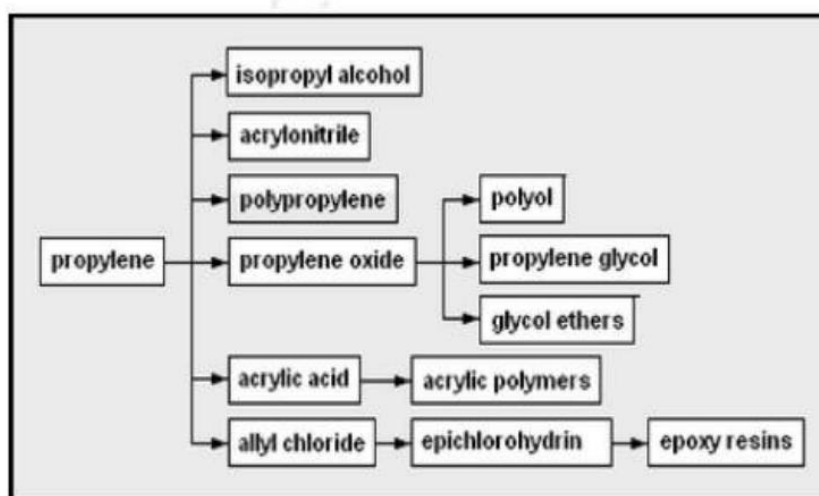


Chemicals produced from ethylene

*propylene:

propylene -used as a monomer and a chemical feedstock

- isopropyl alcohol-2-propanol;often used as a solvent or rubbing alcohol
- acrylonitrile -useful as a monomer in forming Orlon,ABS
- polypropylene -polymerized propylene
- propylene oxide
- polyether polyol -used in the production of polyurethanes
- propylene glycol -used in engine coolant and aircraft deicer fluid
- glycol ethers -from condensation of glycols
- acrylic acid
 - acrylic polymers
 - allyl chloride
 - epichlorohydrin -chloro-oxirane;used in epoxy resin formation
- epoxy resins-a type of polymerizing glue from bisphenol A, epichlorohydrin, and some amine



Chemicals produced from propylene

Aromatics (BTX)

*benzene:-

benzene -the simplest aromatic hydrocarbon

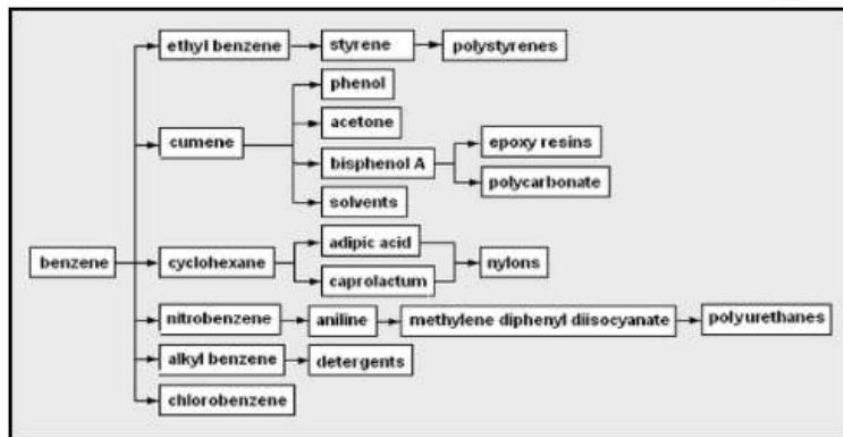
- ethylbenzene -made from benzene and ethylene
 - styrene made by dehydrogenation of ethylbenzene;used as a monomer
 - polystyrenes -polymers with styrene as a monomer
- cumene-isopropylbenzene;a feedstock in the cumene process
 - phenol -hydroxybenzene;often made by the cumene process
 - acetone -dimethyl ketone;also often made by the cumene process
 - bisphenol A-a type of "double"phenol used in polymerization in epoxy resins and making a common type of polycarbonate
 - epoxy resins -a type of polymerizing glue from bisphenol A,epichlorohydrin, and some amine
 - polycarbonate -a plastic polymer made from bisphenol A and phosgene (carbonyl dichloride)
- solvents -liquids used for dissolving materials;examples often made from petrochemicals include ethanol,isopropyl alcohol,acetone,benzene,toluene, xylenes
- cyclohexane -a 6-carbon aliphatic cyclic hydrocarbon sometimes used as a non-polar solvent
 - adipic acid-a 6-carbon dicarboxylic acid,which can be a precursor used as a co-monomer together with a diamine to form an alternating copolymer form of nylon.
 - nylons -types of polyamides,some are alternating copolymers formed from copolymerizing dicarboxylic acid or derivatives with diamines
 - caprolactam -a 6-carbon cyclic amide
 - nylons -types of polyamides,some are from polymerizing caprolactam
- nitrobenzene-can be made by single nitration of benzene
 - aniline -aminobenzene
 - methylene diphenyl diisocyanate (MDI)-used as a co-monomer with diols or polyols to form polyurethanes or with di-or polyamines to form polyureas
- alkylbenzene -a general type of aromatic hydrocarbon,which can be u

sed as a precursor for a sulfonate surfactant (detergent)

detergents -often include surfactants types such as alkylbenzenesulfonates

and nonylphenol ethoxylates

chlorobenzene

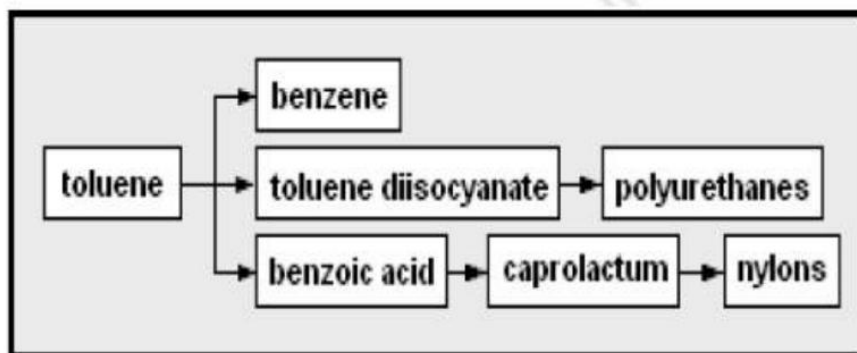


Chemicals produced from benzene

*Toluene

toluene -methylbenzene; can be a solvent or precursor for other chemicals

- benzene
- toluene diisocyanate (TDI)-used as co-monomers with polyether polyols to form polyurethanes or with di- or polyamines to form polyureas polyurethanes
- benzoic acid -carboxybenzene
- caprolactam
- nylon



Chemicals produced from toluene

*xylenes:-

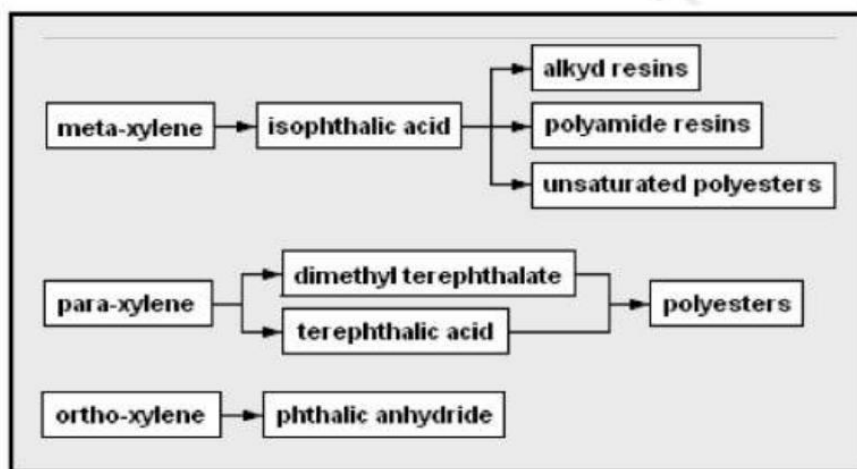
mixed xylenes -any of three dimethylbenzene isomers,could be a solvent but more often

precursor chemicals

- ortho-xylene -both methyl groups can be oxidized to form(ortho-)phthalic acid · phthalicanhydride
- para-xylene -both methyl groups can be oxidized to form terephthalic acid · dimethyl terephthalate -can be copolymerized to form certain polyesters
- polyesters-although there can be many types,polyethylene terep

terephthalate is made from petrochemical products and is very widely used.

- purified terephthalic acid -often copolymerized to form polyethylene terephthalate • polyesters
- *meta-xylene*
 - isophthalic acid
 - alkyd resins
 - Polyamide Resins
 - Unsaturated Polyesters



Chemicals produced from xylenes

Primary Raw Materials for Petrochemicals

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NATURAL GAS

(Non-associated and Associated Natural Gases)

Natural gas is a naturally occurring mixture of light hydrocarbons accompanied by some non-hydrocarbon compounds. Non-associated natural gas is found in reservoirs containing no oil (dry wells). Associated gas, on the other hand, is present in contact with an

d/or dissolved in crude oil and is coproduced with it. The principal component of most

Table 1-1
Composition of non-associated and associated natural gases¹

Component	Non-associated gas		Associated gas	
	Salt Lake US	Kliffside US	Abqaiq Saudi Arabia	North Se UK
Methane	95.0	65.8	62.2	85.9
Ethane	0.8	3.8	15.1	8.1
Propane	0.2	1.7	6.6	2.7
Butanes	—	0.8	2.4	0.9
Pentane and Heavier	—	0.5	1.1	0.3
Hydrogen sulfide	—	—	2.8	—
Carbon dioxide	3.6	—	9.2	1.6
Nitrogen	0.4	25.6	—	0.5
Helium	—	1.8	—	—

natural gases is methane. Higher molecular weight paraffinic hydrocarbons (C_2 – C_7) are usually present in smaller amounts with the natural gas mixture, and their ratios vary considerably from one gas field to another. Non-associated gas normally contains a higher methane ratio than associated gas, while the latter contains a higher ratio of heavier hydrocarbons. Table 1-1 shows the analyses of some selected non-associated and associated gases. In our discussion, both non-associated and associated gases will be referred to as natural gas. However, important differences will be noted.

The non-hydrocarbon constituents in natural gas vary appreciably from one gas field to another. Some of these compounds are weak acids, such as hydrogen sulfide and carbon dioxide. Others are inert, such as nitrogen, helium and argon. Some natural gas reservoirs contain enough helium for commercial production.

Higher molecular weight hydrocarbons present in natural gases are important fuels as well as chemical feedstocks and are normally recovered as natural gas liquids. For example, ethane may be separated for use as a feedstock for steam cracking for the production of ethylene. Propane and butane are recovered from natural gas and sold as liquefied petroleum gas (LPG). Before natural gas is used it must be processed or treated to remove the impurities and to recover the heavier hydrocarbons (heavier than methane). The 1998 U.S. gas consumption was approximately 22.5 trillion ft^3 .

NATURAL GAS TREATMENT PROCESSES

Raw natural gases contain variable amounts of carbon dioxide, hydrogen sulfide, and water vapor. The presence of hydrogen sulfide in natural gas for domestic consumption cannot be tolerated because it is poisonous. It also corrodes metallic equipment. Carbon dioxide is undesirable, because it reduces the heating value of the gas and solidifies under the high pressure and low temperatures used for transporting natural gas. For obtaining a sweet, dry natural gas, acid gases must be removed and water vapor reduced. In addition, natural gas with appreciable amounts of heavy hydrocarbons should be treated for their recovery as natural gas liquids.

Acid Gas Treatment

Acid gases can be reduced or removed by one or more of the following methods:

1. Physical absorption using a selective absorption solvent.
2. Physical adsorption using a solid adsorbent.
3. Chemical absorption where a solvent (a chemical) capable of reacting reversibly with the acid gases is used.

Physical Absorption

Important processes commercially used are the Selexol, the Sulfinol, and the Rectisol processes. In these processes, no chemical reaction occurs between the acid gas and the solvent. The solvent, or absorbent, is a liquid that selectively absorbs the acid gases and leaves out the hydrocarbons. In the Selexol process for example, the solvent is dimethyl ether of polyethylene glycol. Raw natural gas passes countercurrently to the descending solvent. When the solvent becomes saturated with the acid gases, the pressure is reduced, and hydrogen sulfide and carbon dioxide are desorbed. The solvent is then recycled to the absorption tower. Figure 1-1 shows the Selexol process.

Physical Adsorption

In these processes, a solid with a high surface area is used. Molecular sieves (zeolites) are widely used and are capable of adsorbing large amounts of gases. In practice, more than one adsorption bed is used for continuous operation. One bed is in use while the other is being regenerated.

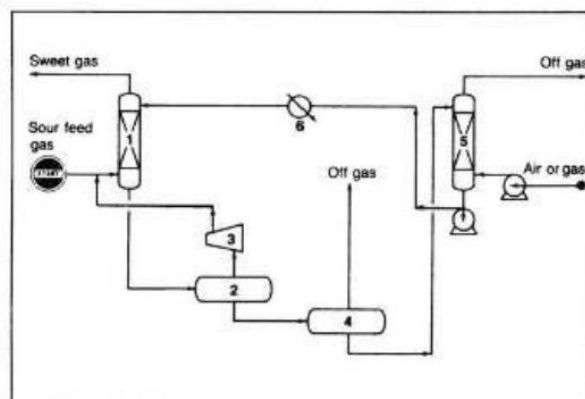


Figure 1-1. The Selexol process for acid gas removal: (1) absorber, (2) flash drum, (3) compressor, (4) low-pressure drum, (5) stripper, (6) cooler.

Regeneration is accomplished by passing hot dry fuel gas through the bed. Molecular sieves are competitive only when the quantities of hydrogen sulfide and carbon disulfide are low.

Molecular sieves are also capable of adsorbing water in addition to the acid gases.

Chemical Absorption (Chemisorption)

These processes are characterized by a high capability of absorbing large amounts of acid gases. They use a solution of a relatively weak base, such as monoethanolamine. The acid gas forms a weak bond with the base which can be regenerated easily. Mono- and diethanolamines are frequently used for this purpose. The amine concentration normally ranges between 15 and 30%. Natural gas is passed through the amine solution where sulfides, carbonates, and bicarbonates are formed.

Diethanolamine is a favored absorbent due to its lower corrosion rate, smaller amine loss potential, fewer utility requirements, and minimal reclaiming needs.³ Diethanolamine also reacts reversibly with 75% of carbonyl sulfides (COS), while the mono- reacts irreversibly with 95% of the COS and forms a degradation product that must be disposed of.

Diglycolamine (DGA), is another amine solvent used in the Econamine process (Fig 1-2).⁴ Absorption of acid gases occurs in an absorber containing an aqueous solution of DGA, and the heated

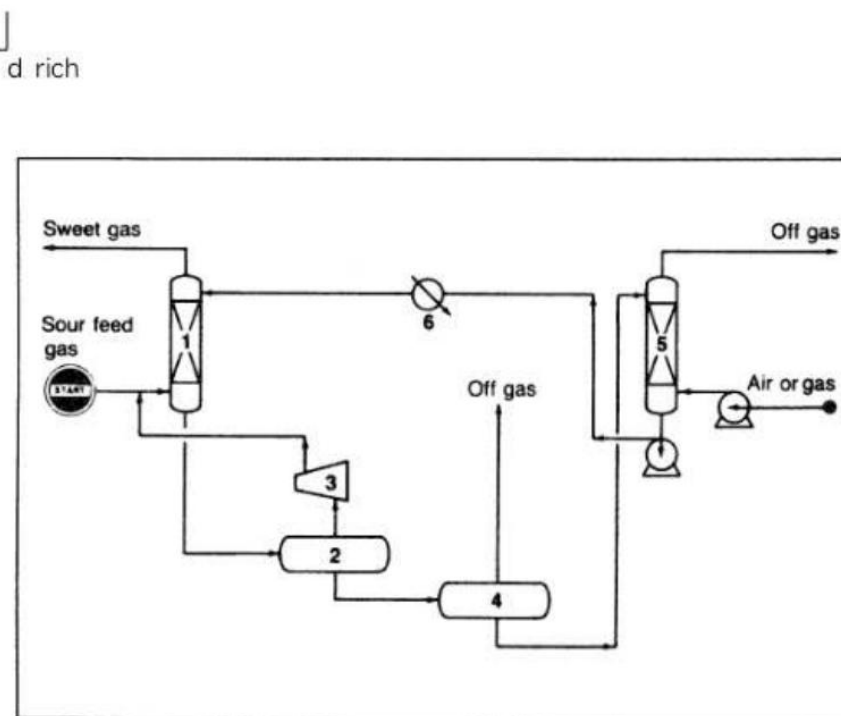
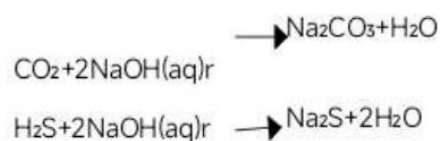


Figure 1-2. The Econamine process: (1) absorption tower, (2) regeneration tower.

solution (saturated with acid gases) is pumped to the regenerator. Diglycolamine solutions are characterized by low freezing points, which make them suitable for use in cold climates.

Strong basic solutions are effective solvents for acid gases. However, these solutions are not normally used for treating large volumes of natural gas because the acid gases form stable salts, which are not easily regenerated. For example, carbon dioxide and hydrogen sulfide react with aqueous sodium hydroxide to yield sodium carbonate and sodium sulfide, respectively.



However, a strong caustic solution is used to remove mercaptans from gas and liquid streams. In the Merox Process, for example, a caustic solvent containing a catalyst such as cobalt, which is capable of converting mercaptans (RSH) to caustic insoluble disulfides (RSSR), is used for streams rich in mercaptans after removal of H_2S . Air is used to oxidize the mercaptans to disulfides. The caustic solution is then recycled for regeneration. The Merox process (Fig. 1-3) is mainly used for treatment of refinery gas streams.

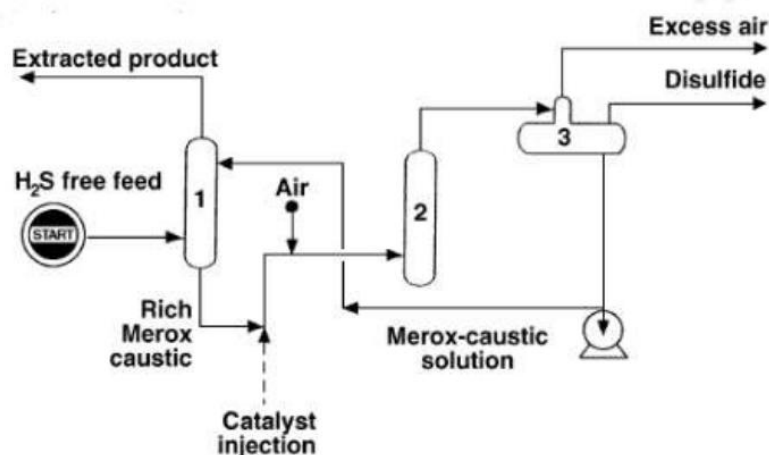


Figure 1-3. The Merox process: (1) extractor, (2) oxidation reactor.

Water Removal

Moisture must be removed from natural gas to reduce corrosion problems and to prevent hydrate formation. Hydrates are solid white compounds formed from a physical-chemical reaction between hydrocarbons and water under the high pressures and low temperatures used to transport natural gas via pipeline. Hydrates reduce pipeline efficiency.

To prevent hydrate formation, natural gas may be treated with glycols, which dissolve water efficiently. Ethylene glycol (EG), diethylene glycol (DEG), and triethylene glycol (TEG) are typical solvents for water removal. Triethylene glycol is preferable in vapor phase processes because of its low vapor pressure, which results

in less glycol loss. The TEG absorber normally contains 6 to 12 bubble-cap trays to accomplish the water absorption. However, more contact stages may be required to reach dew points below -40°F . Calculations to determine the number of trays or feet of packing, the required glycol concentration, or the glycol circulation rate require vapor-liquid equilibrium data. Predicting the interaction between TEG and water vapor in natural gas over a broad range allows the designs for ultra-low dew point applications to be made.⁹

A computer program was developed by Grandhidsan et al., to estimate the number of trays and the circulation rate of lean TEG needed to dry natural gas. It was found that more accurate predictions of the rate could be achieved using this program than using hand calculation.⁹

Figure 1-4 shows the Dehydrate process where EG, DEG, or TEG could be used as an absorbent.⁸ One alternative to using bubble-cap trays

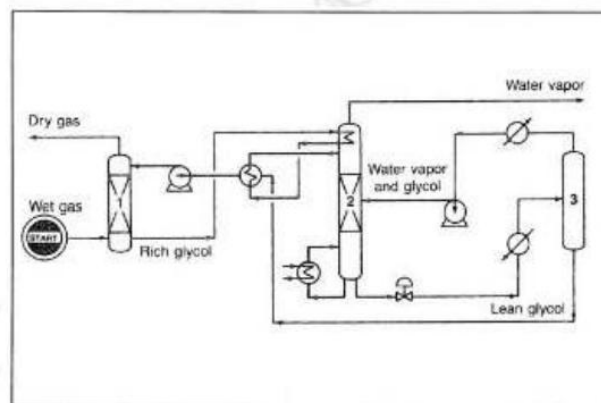


Figure 1-4. Flow diagram of the Dehydrate process⁸: (1) absorption column, (2) glycol still, (3) vacuum drum.

is structural packing, which improves control of mass transfer. Flow passages direct the gas and liquid flows countercurrent to each other. The use of structural packing in TEG operations has been reviewed by Kean et al.⁹ Another way to dehydrate natural gas is by injecting methanol into gas

lines to lower the hydrate-formation temperature below ambient. Water can also be reduced or removed from natural gas by using solid adsorbents such as molecular sieves or silica gel.

Condensable Hydrocarbon Recovery

Hydrocarbons heavier than methane that are present in natural gases are valuable raw materials and important fuels. They can be recovered by lean oil extraction. The first step in this scheme is to cool the treated gas by exchange with liquid propane. The cooled gas is then washed with a cold hydrocarbon liquid, which dissolves most of the condensable hydrocarbons. The uncondensed gas is dry natural gas and is composed mainly of methane with small amounts of ethane and heavier hydrocarbons. The condensed hydrocarbons or natural gas liquids (NGL) are stripped from the rich solvent, which is recycled. Table 1-2 compares the analysis of natural gas before and after treatment. Dry natural gas may then be used either as a fuel or as a chemical feedstock.

Another way to recover NGL is through cryogenic cooling to very low temperatures (-150 to -180°F), which are achieved primarily through

Table 1-2

Typical analysis of natural gas before and after treatment

Component mole %	Feed	Pipeline gas
N ₂	0.45	0.62
CO ₂	27.85	3.50
H ₂ S	0.0013	—
C ₁	70.35	94.85
C ₂	0.83	0.99
C ₃	0.22	0.003
C ₄	0.13	0.004
C ₅	0.06	0.004
C ₆₊	0.11	0.014

adiabatic expansion of the inlet gas. The inlet gas is first treated